EXERCISE

A. Choose the right answer:

1. The device that converts hard copy into soft copy is:
   a) printer b) plotter
   c) scanner d) barcode reader

2. The volatile memory
   a) is permanent
   b) loses contents as the power is disconnected
   c) possesses large storage
   d) manages hardware resources

3. Media players are:
   a) business software b) education software
   c) entertainment software d) productivity software

4. The programs that are generally installed to manage and maintain overall computer resources is:
   a) operating system b) utility program
   c) language translator d) device driver

5. Modern languages use
   a) compiler b) interpreter
   c) convertor d) assembler

6. A collection of wires connecting the CPU with main memory that is used to identify particular locations is:
   a) control bus b) data bus
   c) address bus d) memory bus

7. The inexpensive and most commonly used computers are:
   a) super computer b) mainframe computer
   c) minicomputer d) microcomputer

8. Computer cannot start without:
   a) operating system b) utility program
   c) device drivers d) business software

9. Graphical User Interface (GUI) was developed in:
   a) second generation b) fourth generation
   c) mechanical era d) electro-mechanical era

10. A person who uses different programming languages to develop programs is:
   a) database administrator b) web designer
   c) software engineer d) graphic designer

B. Respond the following:

1. Discuss the use of computer in any two fields of life.
2. Differentiate Compiler and Assembler.
3. Differentiate System and Application software.
4. Describe artificial intelligence with examples.
5. Discuss Impact and Non-Impact Printers with examples.
6. Write the use of these storage devices: Hard Disk, USB Flash Disk, SD Card.
7. Which monitor will you prefer in your school; CRT or FPD? Why?
8. List any five components present on motherboard.
9. Prepare a table of generations.

Generation	Period	Technology	Example of Machines
First			
Second			
Third			
Forth			
Fifth			

- Office Automation Software is a group of computer programs that help office workers in their daily work.
- MS Office 2010 contains MS Word, MS Excel, MS PowerPoint and MS Outlook.
- For MS- Office 2013 the minimum requirements are: 1.6 GHz processor, 1 GB RAM, 3 GB hard disk, minimum Windows 7 operating system and graphics driver.



A. Choose the right answer:

1. The software which performs all basic tasks is:
   a) antivirus b) start menu
   c) operating system d) office automation

2. A program that enables user to interact with computer is called:
   a) my document b) start menu
   c) desktop d) interface

3. The process of checking computer, starting operating system and making it ready to work is referred to as:
   a) resource management b) booting
   c) error checking d) file management

4. The number of user(s) that can access the resources simultaneously on a Multiuser OS is /are:
   a) one b) two
   c) many d) only administrators

5. Which management controls the dynamic allocation and de-allocation of processor, memory, etc?
   a) Resource b) File
   c) I/O d) User

6. In which operating system jobs are executed in groups?
   a) Batch Processing b) Time Sharing
   c) Single User d) Real Time

7. The system that guarantees that critical tasks should be completed without shortest delay is:
   a) Batch Processing
   b) Hard Real Time
   c) Soft Real Time
   d) Time Sharing

8. DOS is an example of:
   a) Real Time Processing OS
   b) Multi Processing OS
   c) Single User OS
   d) Multi User OS

9. The group of programs that helps office workers to do a number of routine tasks easily and efficiently is called:
   a) operating system
   b) e-mail
   c) anti-virus
   d) office automation

10. The operating system that allows frequent switching from one task to another is:
   a) Batch Processing
   b) Real Time Processing
   c) Single User
   d) Time Sharing

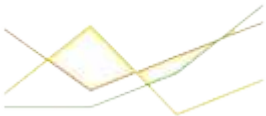
B. Respond the following:

1. Define Operating System. Give three examples of operating systems.
2. List the functions of an OS? Write briefly about any two.
3. Differentiate between CLI and GUI. Write any two benefits of each.
4. Which resources are managed by Resource Management function of OS?
5. What is the difference between single user and multi user OS? If you are a manager of a large organization which type of OS will you prefer? Justify your answer with any two reasons.
6. What is the purpose of office automation software? With examples, explain how can office automation help employees to be productive and efficient?
7. List the programs available in MS Office and write why each program is used?
8. What is the major difference between Soft Real Time and Hard Real Time systems?
9. Why it is important to install an antivirus?

EXERCISE

A. Choose the right answer:

1. Which chart will be suitable to show the share of three partners in a business?

a) 

b) 

c) 

d) 

2. The software that is used for accounting purpose is:
   a) MS Word b) MS Excel
   c) MS Power point d) MS Access

3. The software that is used to format a document is:
   a) MS Word b) MS Excel
   c) MS Power point d) MS Access

4. The special character that initiates the formula mode in a cell is:
   a) / b) =
   c) - d) *

5. The function which is used to re-arrange data according to specific criteria is called:
   a) Filtering b) Sorting
   c) Organizing d) Grouping

6. If we want to change the overall design of a word document, we should use:
   a) themes b) page layout
   c) watermark d) margins

7. The correct formula to calculate the total in the given table is:

	A	B	C	D	E	F
1	Name	English	Sindh/Urdu	Maths	Science	Total
2	Ghulam Shabir	40	41	41	40	
3	Riaz Hussain	41	39	34	38	
4	Nabil Ahmed	41	35	32	40	
5	Anwar Ali	40	38	33	37	

a) = B2 + C2 + D2+ E2 b) = B2 + E2
c) = sum (B2 to E2) d) = sum (B2 from E2)

8. The correct formula to calculate the percentage in the given table is:

	A	B	C	D	E	F
1	Name	English	Sindh/Urdu	Maths	Science	Per
2	Ghulam Shabir	40	41	41	40	
3	Riaz Hussain	41	39	34	38	
4	Nabil Ahmed	41	35	32	40	
5	Anwar Ali	40	38	33	37	

a) =B2 + C2 + D2+ E2 / 400 * 100
b) =B2 + C2 + D2+ E2 * 100
c) =(B2 + C2 + D2+ E2) / 400 * 100
d) =(B2 + C2 + D2+ E2) / 100 * 400

9. Charts are basically used to
a) design tables
b) organize data in tables
c) Sort data in different columns
d) show the quantities and their relationships graphically

10. To apply the table of contents automatically, the most important task is to properly define the
a) levels of heading b) page numbering
c) page layout d) themes

B. Respond the following:

1. What is a word processor? Write any three applications of a word processor.
2. Discuss the Margins and Paper Size options in the Page Setup group.
3. Name and describe two options in Orientation.
4. How can we apply Margins in a document?
5. Describe the different types of breaks in MS Word.
6. Differentiate “Bring Forward” and “Send Backward” options in the Arrange group with example.

EXERCISE

A. Choose the right answer:

1. Wired Media is also called:
   a) targeted media b) directed media
   c) guided media d) unguided media

2. Communication system is made up of
   a) three components b) four components
   c) five components d) six components

3. Both Physical and Logical addresses are:
   a) different b) unique
   c) permanent d) temporary

4. If you are an electrical or electronic engineer, you should join:
   a) IEEE b) IETF
   c) ITU d) ANSI

5. The topology in which all computers are connected to a central device called hub is:
   a) Bus b) Star
   c) Ring d) Tree

6. Change in the shape of signal between sender and receiver is called:
   a) attenuation b) interruption
   c) noise d) distortion

7. Router determines data path to transfer data packets which is the:
   a) shortest b) longest
   c) cheapest d) optimal

8. Converting digital signal to analog is called:

a) modulation b) modification

c) bandwidth d) multiplexing

9. The number of bits used in an IPV 4 address are:

a) 16 b) 32

c) 64 d) 128

10. The loss of energy in transmission signal refers to.

a) Attenuation b) Distortion

c) Noise d) Jitter

B. Respond the following:

1. List the properties of a good communication system. Explain any one.
2. Explain components of communication using single example.
3. Write the function of following network devices.
   Amplifiers, Routers, Switch, Hub
4. List the causes of signal impairments. Explain any one.
5. What is the difference between radio wave and microwave?
6. Why OSI model is broken up in layers?
7. Explain the purpose of Standard Organization.
8. List one merit and one demerit of each topology.
9. Give one example of LAN, WAN and MAN.
10. How can we measure bit rate and baud rate? Give an example of each.

- Intellectual property is intangible creations of the human intellect. To prevent theft or illegal use or spread of intellectual property, Intellectual Property Right is exercised. Through these rights, intellectual property is protected with the help of copyrights, patents, and trademarks
- Software piracy is the illegal use, copying or distribution of copyrighted software.
- Plagiarism is presenting someone else's work or ideas without full acknowledgment of the author or conceiver.



A. Choose the right answer:

1. The broad term that encompasses different harmful software is:
   a) Virus b) Malware
   c) Adware d) Spyware

2. The authentication mechanism that only allows the real person to access data or device is:
   a) Username and Password b) PIN
   c) Biometric d) Scan Code

3. Software are mostly protected under:
   a) Patents b) Copyrights
   c) Trademarks d) Logos

4. The professional ethics in computer field is important because:
   a) It is necessary by law.
   b) Violation can cause serious penalties.
   c) It is useful for financial benefits.
   d) It creates healthy and positive work environment.

5. Free Antivirus Software often
   a) Expire after sometimes b) Offer only limited service
   c) Cannot be updated d) Cannot be purchased

6. Copying and pasting some texts from internet without acknowledging the real author is an example of:
   a) Plagiarism
   b) Illegal use of patent
   c) Information Piracy
   d) Copyright Violation

7. Since it does not harm or steal data, the least harmful malware is:
   a) Virus
   b) Adware
   c) Spyware
   d) Trojan

8. The malware that replicates itself and doesn't need to attach with any files is:
   a) Virus
   b) Adware
   c) Spyware
   d) Worm

9. Through which virus spreads?
   a) Email Attachments
   b) Internet Downloads
   c) Flash Disks and Cds
   d) All of them

10. "Click this link and win a $5 voucher at McDonald's". This is an example of:
   a) Scam
   b) Phishing
   c) Clickjacking
   d) Hacking

B. Respond the following:

1. Why is computer security important? Write any three reasons.
2. Explain Cyber Bullying with an example.
3. Why is information accuracy important?
4. What is Ethical Hacking?
5. Your friend has become a victim of cyber harassment. What two advices will you give him or her?
6. Write any two measures to avoid email account hacking.
7. How is software piracy harmful for software developers?
8. Give two examples of phishing.
9. What is an Intellectual Property Right?

- Hyperlinks give the ability to any HTML object of being clickable. Hyperlinks provide means of navigation from one page to another or from one section in a page to another within the same page.
- HTML tables can be used to display data, information or content in a tabular format. Tables organize the content into rows and columns.
- HTML Frames can be used to display the entire contents of another web page within a web page.
- Framesets define the layout to display different frames and their contents.
- There are different tools available which facilitate the designing and development of web pages and websites.



A. Choose the right answer:

1. The service that is responsible for making websites publicly accessible through the internet is called

   a) Web Server b) Web Hosting
   c) Web Site d) Web Browser

2. The type of special website where different users can ask questions and give answers or discuss on various topics is called:

   a) Social site b) Blogs site
   c) Forums site d) Informational site

3. Entertainment site specializes in delivering:

   a) News, weather and current affairs
   b) Information about products and services of a business
   c) Personal information of a particular person
   d) Content like videos, images and games for entertainment

4. A web browser will translate the codes in a web page if the extension of the document is:

a) .html
b) .txt
c) .doc
d) .pdf

5. The tag used to automatically add line space before and after the containing text is:

a) `<br>`
b) `<hr>`
c) `<p>`
d) `<pre>`

6. The type of list which shows its items in a numerically ordered sequence is:

a) Nested List
b) Unordered List
c) Ordered List
d) Description List

7. To create a clickable text which navigates to another page or section, we use:

a) `<input>` tag
b) `<li>` tag
c) `<b>` tag
d) `<a>` tag

8. To differentiate the heading cells from rest of the data in a table, we use:

a) `<th>` tag
b) `<tr>` tag
c) `<td>` tag
d) `<dt>` tag

9. The tag used to define the title caption of the web browser is:

a) `<thead>`
b) `<head>`
c) `<title>`
d) `<h1>`

10. The attribute used to define the URL for reference of image in <img> tag is:

a) target
b) name
c) src
d) href

EXERCISE

A. Choose the right answer.

1. Which of the following Microsoft Office packages is a DBMS?
   a) MS- Word b) MS- Excel
   c) MS- Power Point d) MS- Access

2. The basic limitation of a flat file database is that:
   a) It is complicated b) It stores data in a single file
   c) It is very heavy d) It is not supported on internet

3. In a database table of "Students", the address of the student will be a:
   a) Record b) Field
   c) Entity d) Data type

4. In a database table of "Students", the particulars of a single student will be a:
   a) Record b) Field
   c) Entity d) Data type

5. In a relational database, keys are used to create a:
   a) Table b) Fields
   c) Records d) Relationship

6. A field that stores the names of students should be defined as:
   a) Integer b) Float
   c) String d) Boolean

7. A key that allows only unique entries in a field is called:
   a) Primary Key b) Secondary Key
   c) Foreign Key d) Super Key

8. Data Redundancy means:
   a) Duplication of Data b) Variety of Data
   c) Size of Data d) Data Type

9. The relationship that matches one record of an entity with only one record of another entity is called:

a) One-to-One relationship
b) One-to-Many relationship
c) Many-to-One relationship
d) Many-to-Many relationship

10. The shape that is used to represent an attribute in an ERD is:

a) A diamond
b) An octagon
c) A rectangle
d) An ellipse

B. Respond the following:

1. Why Database Management System is preferred over Flat File System?
2. Discuss the use of databases in business with example.
3. What is the difference between Design View and Datasheet View?
4. Why is it important to carefully decide the data type for each field?
5. What is the difference between table and view?
6. Explain entities, attributes and relationships with one example of each.
7. List the major characteristics of primary and foreign keys.
8. Write three benefits of using relationships in the database.
9. Discuss the importance of ERD in business
10. Write any two statements from the following ER Diagram.

